

Linear Regression Model Report

California Housing Price Prediction

Maincrafts Internship Task 1

1 Objective

This report documents the end-to-end machine learning workflow used to build and evaluate a Linear Regression model that predicts median house value for California housing blocks. The workflow covers data loading, exploratory data analysis (EDA), preprocessing, model training, evaluation, and recommendations for improvement.

2 Dataset

The California Housing dataset (Pace & Barry, 1997) contains 20,640 observations describing housing blocks across California, sourced from the 1990 U.S. Census. Each row represents a census block group with the following 8 predictor features and 1 target variable:

- **MedInc** – median income of the block (in \$10,000s)
- **HouseAge** – median age of houses in the block
- **AveRooms** – average number of rooms per household
- **AveBedrms** – average number of bedrooms per household
- **Population** – block population
- **AveOccup** – average number of household members
- **Latitude, Longitude** – block geographic coordinates
- **MedHouseVal** (target) – median house value, in \$100,000 units

The dataset has no missing values, so no imputation was required.

3 Exploratory Data Analysis

Feature distributions were examined to understand scale, skew, and potential outliers before modeling.

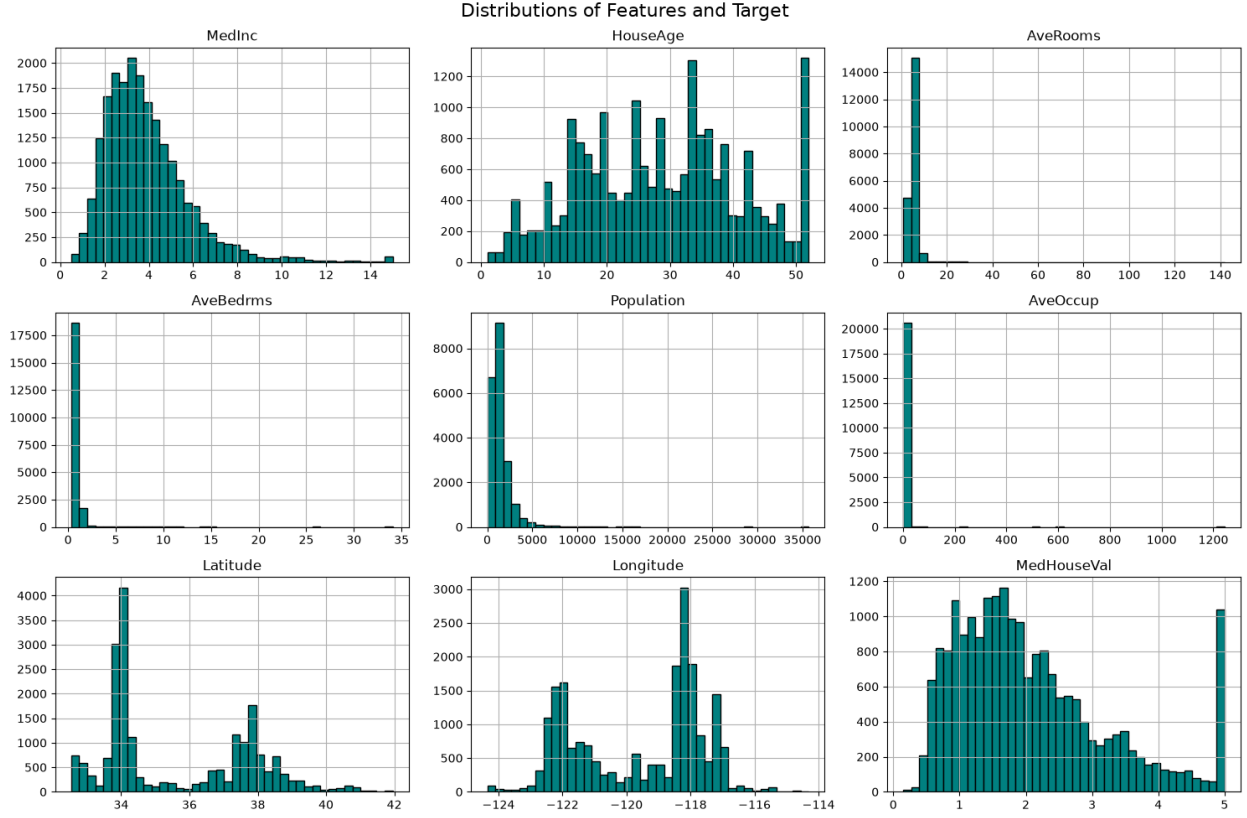


Figure 1: Distributions of all features and the target variable. **MedInc**, **Population**, and **AveOccup** are right-skewed with long tails. **AveRooms** and **AveBedrms** show extreme outliers in a small number of blocks. **MedHouseVal** shows a visible cap near \$500,000, a known artifact of the original census data collection.

A correlation matrix was computed to identify linear relationships between features and the target, and to flag multicollinearity risks.

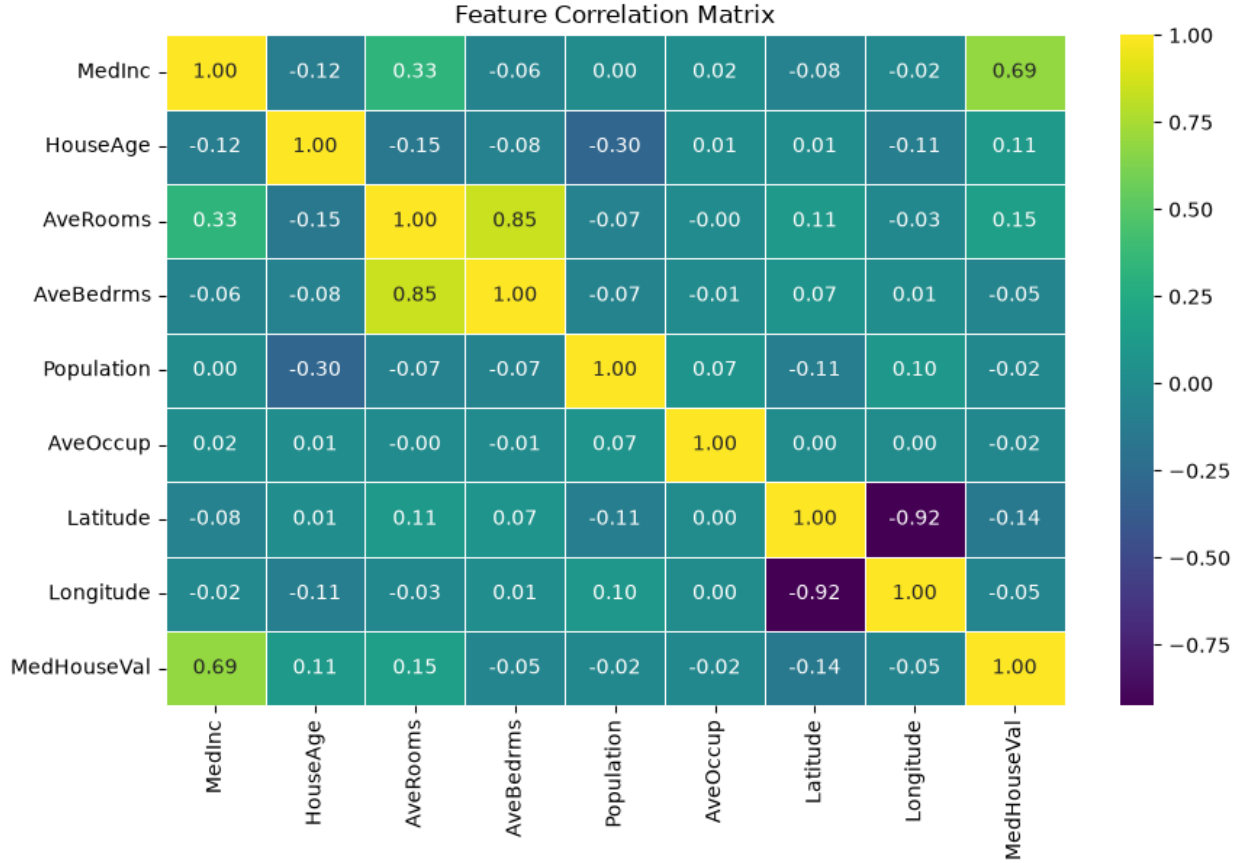


Figure 2: Feature correlation matrix. **MedInc** is the strongest positive predictor of **MedHouseVal** ($r = 0.69$). **Latitude** and **Longitude** are strongly anti-correlated ($r = -0.92$), a geographic artifact of California’s coastline orientation rather than a modeling concern. **AveRooms** and **AveBedrms** are highly correlated with each other ($r = 0.85$), indicating possible redundancy.

4 Preprocessing

The following preprocessing steps were applied prior to training:

1. **Train/test split:** 75% train, 25% test, with `random.state=7` for reproducibility.
2. **Feature scaling:** `RobustScaler` was used instead of standard z-score scaling, since it centers on the median and scales by the interquartile range, making it less sensitive to the outliers observed in **AveRooms**, **AveBedrms**, and **Population**.

No feature selection or dimensionality reduction was applied; all 8 features were retained.

5 Model

A standard `LinearRegression` model from scikit-learn was trained on the scaled training features to predict **MedHouseVal**. Linear regression was chosen as a baseline model for its interpretability and low computational cost, providing a benchmark against which more complex models can later be compared.

6 Evaluation

Model performance was assessed on the held-out 25% test set using three standard regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

Metric	Value	Interpretation
MAE	0.5310	Average prediction is off by \$53,100
RMSE	0.7237	Penalizes larger errors more heavily than MAE
R^2	0.6070	Model explains 60.7% of variance in house values

Table 1: Test set performance metrics.

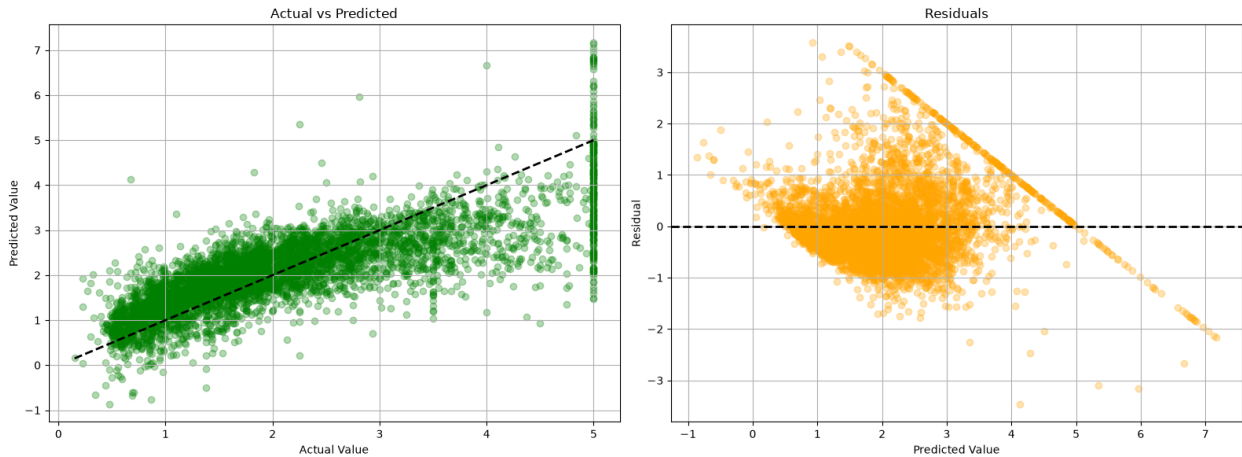


Figure 3: Left: Actual vs. predicted house values, with the dashed line representing perfect prediction. Points cluster reasonably around the line for mid-range values but diverge at the extremes. Right: Residuals plotted against predicted values. The diagonal band of points near \$500k corresponds to the capped target values, which the model cannot predict correctly by construction, systematically inflating error at the top of the value range.

7 Limitations

- The capped target value (\$500,000 ceiling) introduces a hard-to-model discontinuity, visible as the diagonal residual band.
- Linear regression assumes a linear relationship between features and target; the diverging scatter at both extremes suggests underlying non-linear effects.
- Geographic features (**Latitude**, **Longitude**) are treated independently, but house prices are more likely driven by proximity to specific hubs (coast, cities) rather than raw coordinates.

8 Improvement Ideas

- **Non-linear models:** Random Forest or Gradient Boosting Regressors typically outperform linear models on this dataset by capturing non-linear feature interactions.
- **Feature engineering:** derive `rooms_per_household`, `bedrooms_per_room`, and `distance-to-coast` or `distance-to-major-city` features from `Latitude/Longitude`.
- **Transform skewed features:** apply log-transforms to `Population` and `AveOccup` to reduce the influence of long tails.
- **Handle capped targets:** remove or separately model the capped \$500k observations to reduce their distortion of residuals.
- **Cross-validation:** replace the single train/test split with k-fold cross-validation for a more robust performance estimate.
- **Regularization:** try Ridge or Lasso regression to address multicollinearity between `AveRooms` and `AveBedrms`.

9 Conclusion

The baseline Linear Regression model achieved an R^2 of 0.607 on the test set, explaining roughly 61% of the variance in California house values with an average error of about \$53,100 (MAE). This confirms that a simple linear model captures a meaningful portion of the price signal, driven primarily by `MedInc`, but leaves clear room for improvement, especially at the extremes of the value range where the capped target and non-linear effects reduce accuracy. The model serves as a solid, interpretable baseline; the improvement ideas above (non-linear models, feature engineering, and target capping treatment) are the natural next steps toward a production-ready predictor.